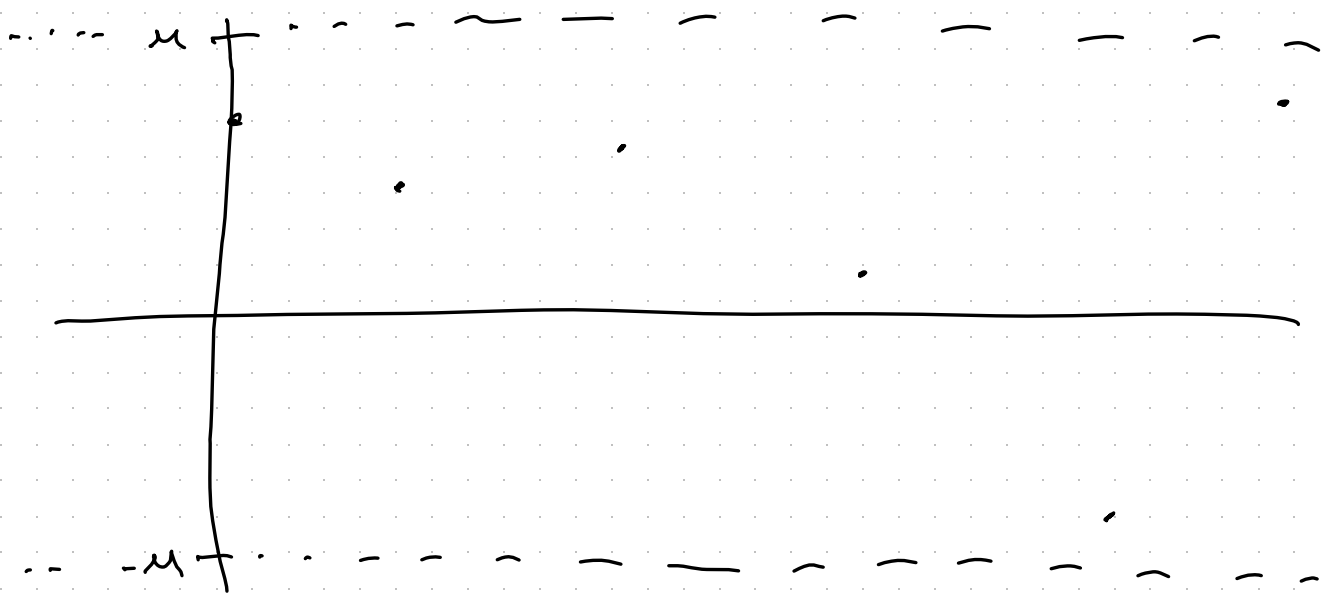
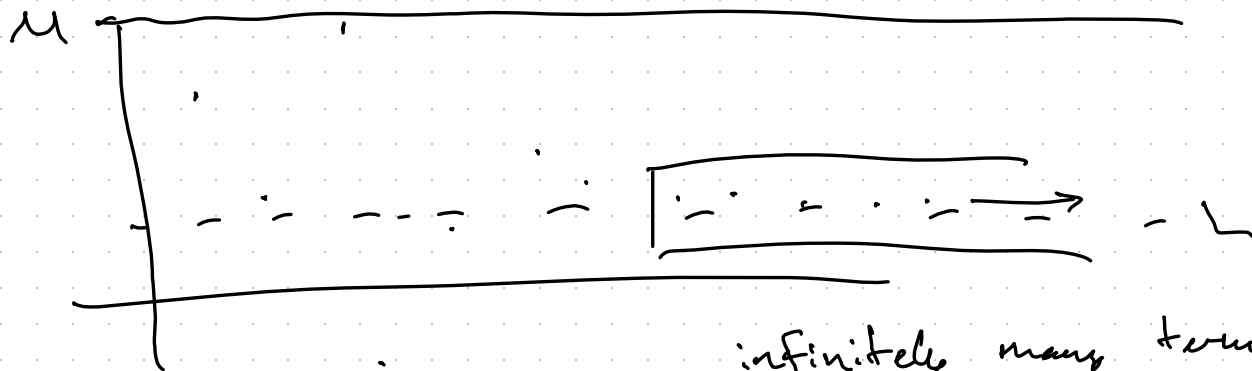


Monotone Convergence Theorem - 5.1

Definition: A sequence (a_n) is bounded if there exists an M such that $|a_n| \leq M$.



Proposition: Any convergent sequence is bounded.

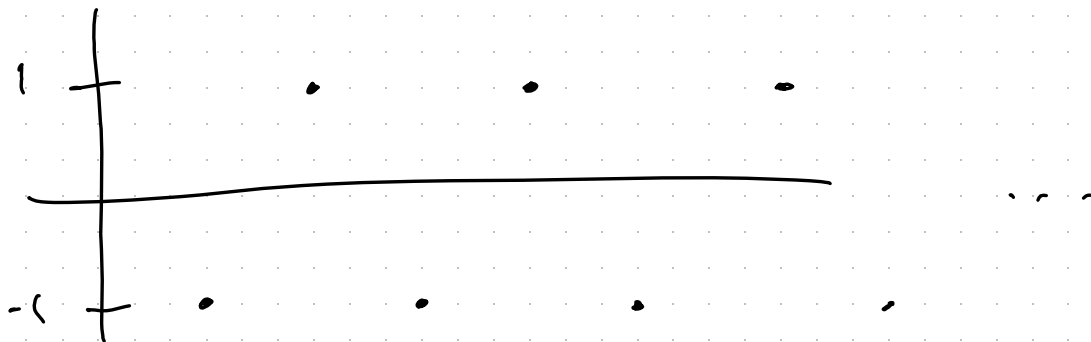


Finitely many
terms are
arbitrarily large

infinitely many terms
are very close to L

Not all bounded sequences converge:

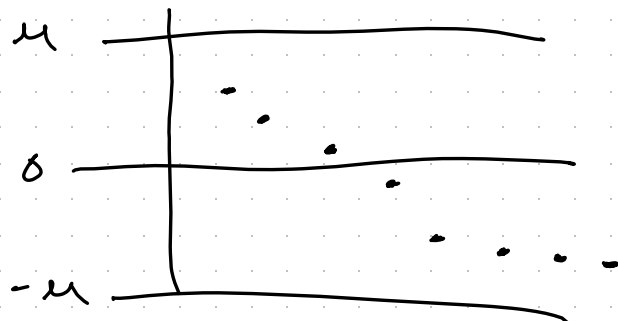
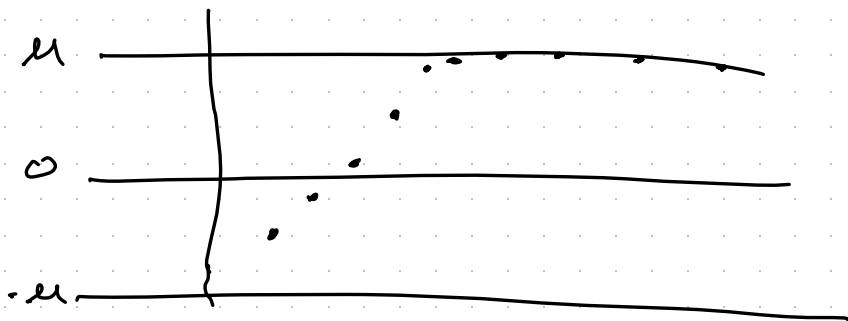
$a_n = (-1)^n$ is bounded ($M=1$) but
does not converge



Definition: A sequence (a_n) is monotone if for all n , $a_{n+1} \geq a_n$ or for all n , $a_{n+1} \leq a_n$.

(an increasing sequence or a decreasing one)

Monotone Convergence Theorem: If (a_n) is bounded and monotone, then it converges.



Consequence: We defined a^x for real x
essentially used the MCT:

→ we said a^x makes sense for
rational numbers x , and we constructed
an increasing sequence with limit x

(e.g. $3, 3.1, 3.14, 3.141, \dots$)

and plugged this into the function to get
a new increasing / decreasing sequence
 $a \geq 1$ $a \leq 1$

($a^3, a^{3.1}, a^{3.14}, a^{3.141}, \dots$) and the MCT

says this new sequence converges. (call the limit
 a^x)

Sometimes we can also use the MCT to compute a limit.

Example: Find the limit of $(\sqrt{2}, \sqrt{2\sqrt{2}}, \sqrt{2\sqrt{2\sqrt{2}}}, \dots)$ if it exists.

Observation: for $0 \leq x \leq 2$, $0 \leq \sqrt{x} \leq \sqrt{2}$,

$$\text{so } x = \sqrt{x} \cdot \sqrt{x} \leq \sqrt{2} \cdot \sqrt{x} \leq \sqrt{2} \cdot \sqrt{2} = 2,$$

thus $x \leq \sqrt{2x} \leq 2$, so if we start with

$\sqrt{2}$ and then define $a_{n+1} = \sqrt{2a_n}$, then

all values of the sequence are in $[0, 2]$.

→ bounded and increasing.

We can apply the MCT to say
that $a_1 = \sqrt{2}$, $a_{n+1} = \sqrt{2a_n}$ does converge.

The limit L does exist: $a_n \rightarrow L$.

Then also $\frac{a_n^2}{2} \rightarrow \frac{L^2}{2}$

$$\frac{a_n^2}{2} : 1, \sqrt{2}, \sqrt{2\sqrt{2}}, \sqrt{2\sqrt{2\sqrt{2}}}, \dots \rightarrow L$$

Conclusion: this number L satisfies $\frac{L^2}{2} = L$

$$L^2 = 2L, \quad L^2 - 2L = 0, \quad L(L-2) = 0$$

$$L \neq 0, \quad \text{so} \quad L = 2.$$